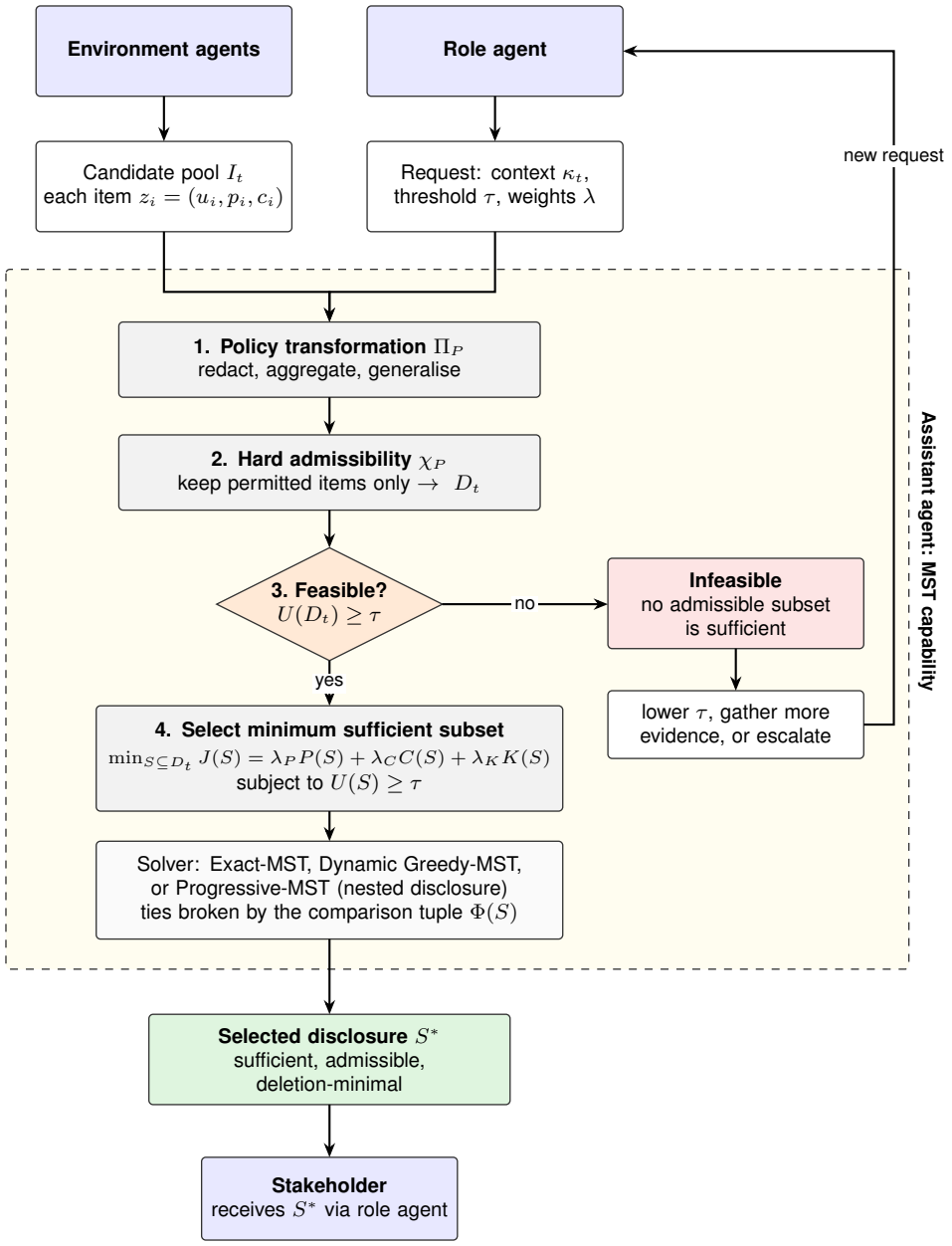

Environment agents
Role agent
Candidate pool I_t
each item $z_i = (u_i, p_i, c_i)$
Request: context κ_t,
threshold τ, weights λ
new request
Assistant agent: MST capability
1. Policy transformation Π_P
redact, aggregate, generalise
2. Hard admissibility χ_P
keep permitted items only $\rightarrow$ D_t
3. Feasible?
$U(D_t) \geq \tau$
no
Infeasible
no admissible subset
is sufficient
lower τ, gather more
evidence, or escalate
yes
4. Select minimum sufficient subset
$\min_{S \subseteq D_t} J(S) = \lambda_P P(S) + \lambda_C C(S) + \lambda_K K(S)$
subject to $U(S) \geq \tau$
Solver: Exact-MST, Dynamic Greedy-MST,
or Progressive-MST (nested disclosure)
ties broken by the comparison tuple $\Phi(S)$
Selected disclosure S^*
sufficient, admissible,
deletion-minimal
Stakeholder
receives S^* via role agent